

MChimanal

Automated Dual-Channel Fluorescence Confocal Image Analysis Workflow for Fiji/ImageJ (1.54p)

Detailed description of the code's functionality (pseudocode)

Implementation note. In the dual-channel workflow, C2 was used as the reference channel for particle segmentation and threshold-method evaluation. The intensity range determined from C2 was applied to both C2 and C1 before conversion to 8-bit format and subsequent preprocessing. Particle detection, particle-size statistics, and threshold-method consistency analysis were performed on C2, whereas both channels were used for the subsequent colocalization analysis.

Thresholding-method consistency was assessed on the boundary-excluded C2 particle measurements using a Shewhart-inspired criterion rather than a classical Shewhart control chart. For each comparison metric, the median across the nine thresholding methods was used as the central reference value, while the sample standard deviation across the same nine values was used as the measure of dispersion. A thresholding method was assigned to the 1 SD, 2 SD, or 3 SD consistency category only when all four evaluated quantities—median particle area, median perimeter-derived diameter, median area-equivalent circular diameter, and particle count—simultaneously satisfied the corresponding criterion.

Particle measurements were performed in two modes: with objects touching the image boundary excluded (boundary-excluded analysis) and with such objects retained (boundary-inclusive analysis). Spatial units were inherited from the calibration recognized by Fiji/ImageJ for the input microscopy image.

Colocalization analysis was performed separately on the original dual-channel images using BIOP JACoP and a thresholding method taken from the exported threshold-method consistency classification. Diagnostic plots, fluorograms, segmentation overlays, intermediate data files, and organized output directories generated by the software were used for workflow inspection, quality control, and reproducibility and should be distinguished from the quantitative measurements and summary statistics used for downstream analysis.

Algorithm 1. Automated dual-channel fluorescence image analysis

INPUT:

- Dual-channel confocal fluorescence images
- Set of nine automatic thresholding methods:
 - Default
 - Huang
 - Intermodes
 - IsoData
 - Li
 - MaxEntropy
 - Moments
 - Otsu
 - RenyiEntropy

OUTPUT:

- C2 particle measurements obtained with each thresholding method
- C2 particle-size statistics
- Fluorescence-intensity distributions for C1 and C2
- C2 segmentation overlays
- Comparison of thresholding methods
- Dual-channel colocalization measurements

1. Scan the working directory and identify LSM images designated for analysis according to the configured filename convention.
2. FOR each selected input image:
 - a. Load the image.
 - b. Convert the image to 16-bit format if required.
 - c. Split the image into individual channels.
 - d. Identify channels C1 and C2 from their image names.
 - IF C1 or C2 cannot be identified from their names:
 - IF at least two channels are present:
 - assign the first available channel as C1
 - assign the second available channel as C2
 - ELSE:
 - terminate analysis of the image because two channels are required.
 - e. Use C2 as the reference channel.
 - f. Determine the C2 intensity-display range:
 - obtain the C2 intensity histogram
 - identify the first and last non-empty histogram bins
 - constrain the lower and upper limits away from the outer 0.2% of the histogram-bin range
 - calculate the corresponding lower and upper intensity values.
 - g. Preprocess C2:
 - apply the C2-derived intensity range
 - apply the LUT
 - convert to 8-bit with intensity scaling
 - apply Gaussian blur with sigma = 2
 - apply smoothing
 - apply despeckling
 - save the preprocessed C2 image
 - calculate and save the C2 fluorescence-intensity histogram
 - create a working copy of preprocessed C2.
 - h. Preprocess C1 using the same reference intensity range derived from C2:
 - apply the C2-derived intensity range
 - apply the LUT
 - convert to 8-bit with intensity scaling
 - apply Gaussian blur with sigma = 2
 - apply smoothing
 - apply despeckling
 - save the preprocessed C1 image
 - calculate and save the C1 fluorescence-intensity histogram.
 - i. FOR each of the nine automatic thresholding methods:
 - i. Duplicate the preprocessed C2 working image.
 - ii. Determine the intensity threshold automatically using the selected thresholding method.
 - iii. Convert the thresholded C2 image to a binary mask.
 - iv. Apply watershed segmentation to separate adjacent particles.
 - v. Perform particle analysis using a lower particle-area limit of 0.1 in the area units recognized by Fiji/ImageJ for the input image.
 - vi. Perform boundary-excluded analysis:
 - exclude particles touching the image boundary

- measure the remaining particles
 - save particle measurements
 - save the labelled segmentation overlay.
- vii. Perform boundary-inclusive analysis:
 - retain particles touching the image boundary
 - measure all detected particles
 - save particle measurements
 - save the labelled segmentation overlay.
- viii. Close the temporary segmented C2 image.

END FOR

- j. Calculate C2 particle-size statistics independently for each thresholding method.
- k. Evaluate the consistency of the boundary-excluded C2 particle measurements across the nine thresholding methods using the Shewhart-inspired consistency criterion described in Algorithm 3.
 - l. Perform dual-channel colocalization analysis as described in Algorithm 4.
 - m. Export and organize the analysis outputs as described in Algorithm 5.

END FOR

Algorithm 2. Calculation of C2 particle-size statistics

INPUT:

C2 particle measurements obtained for one automatic thresholding method

FOR each detected particle:

1. Obtain the projected particle area, A.
2. Obtain the particle perimeter, P.
3. Determine the perimeter-derived diameter: $D_P = P / \pi$
4. Determine the area-equivalent circular diameter: $D_A = 2 * \sqrt{A / \pi}$

END FOR

FOR each thresholding method calculate:

- median particle area
- first quartile (Q1) of particle area
- third quartile (Q3) of particle area
- median perimeter-derived diameter
- Q1 of perimeter-derived diameter
- Q3 of perimeter-derived diameter
- median area-equivalent circular diameter
- Q1 of area-equivalent circular diameter
- Q3 of area-equivalent circular diameter
- total number of detected particles
- square root of the particle count as an auxiliary output quantity.

END FOR

Perform these calculations independently for:

boundary-excluded analysis

and

boundary-inclusive analysis.

Export the particle-level measurements and per-method summary statistics.

Algorithm 3. Shewhart-inspired consistency criterion for C2 threshold-method classification

INPUT:

Boundary-excluded C2 particle statistics obtained independently using the nine automatic thresholding methods

1. For each thresholding method obtain:
 - median particle area
 - median perimeter-derived diameter
 - median area-equivalent circular diameter
 - number of detected particles.
 2. For each of the four quantities calculate the central reference value across the nine thresholding methods:
 - reference particle area = median of the nine model-specific median particle areas
 - reference perimeter-derived diameter = median of the nine model-specific median perimeter-derived diameters
 - reference area-equivalent circular diameter = median of the nine model-specific median area-equivalent circular diameters
 - reference particle count = median of the nine model-specific particle counts.
 3. Calculate the sample standard deviation across the nine thresholding methods separately for:
 - median particle area
 - median perimeter-derived diameter
 - median area-equivalent circular diameter
 - particle count.
 4. FOR each thresholding method:
 - calculate its absolute deviation from the corresponding across-method median for each of the four quantities.
 - IF all four quantities are within 1 SD of their respective across-method medians:
 - classify the thresholding method as "Within 1 SD".
 - ELSE IF all four quantities are within 2 SD of their respective across-method medians:
 - classify the thresholding method as "Within 2 SD".
 - ELSE IF all four quantities are within 3 SD of their respective across-method medians:
 - classify the thresholding method as "Within 3 SD".
 - ELSE:
 - classify the thresholding method as "Outside 3 SD".
- END FOR
5. Identify thresholding methods fulfilling the predefined Shewhart-inspired consistency criterion for subsequent quantitative analysis.
 6. Export the across-method reference statistics and the thresholding methods classified as Within 1 SD, Within 2 SD, or Within 3 SD, together with their consistency categories.

Algorithm 4. Dual-channel colocalization analysis

INPUT:

Original dual-channel fluorescence image
Exported threshold-method consistency classification

1. Determine the automatic thresholding method to be used for colocalization:
 - IF the threshold-method classification file exists:
 - read the first thresholding method listed in the exported classification table
 - ELSE:
 - use the Default thresholding method.
2. Reopen the original fluorescence image.
3. Convert the image to 16-bit format if required.

4. Split the image into channels, identify channels whose names contain C1/Ch1 or C2/Ch2, rename them C1 and C2, respectively, and close other channels.
5. Generate and save a merged two-channel image for visualization.
6. Run BIOP JACoP using the two fluorescence channels:
 - use the selected automatic thresholding method for both channels
 - use automatic rather than manually specified threshold values
 - calculate the requested colocalization metrics, including:
 - Pearson's correlation coefficient
 - Spearman's rank correlation coefficient
 - Manders M1 and M2 coefficients
 - thresholded Manders M1 and M2 coefficients
 - overlap coefficients
 - Li's intensity correlation quotient (ICQ)
 - cytofluorogram parameters
 - available channel-area and overlap-area measurements.
 - generate a fluorogram.
7. Save the initial JACoP output as variant VAR0 using:
 - 256 fluorogram bins
 - fluorogram stroke width = 0.2.
8. Evaluate the graphical quality of the VAR0 fluorogram:
 - calculate a gridiness score from the fraction of image rows and columns having intensity standard deviation below 8.
 - $\text{gridiness} = (\text{fraction of low-variation columns} + \text{fraction of low-variation rows}) / 2.$
9. IF the VAR0 fluorogram is missing OR its gridiness score is greater than 0.25:
 - rerun the JACoP analysis as variant VAR1 using:
 - 128 fluorogram bins
 - fluorogram stroke width = 0.
 - calculate the VAR1 gridiness score.
10. IF VAR1 was generated AND its gridiness score is lower than the VAR0 gridiness score:
 - select VAR1 as the output variant
- ELSE:
 - retain VAR0 as the output variant.
11. Parse the colocalization measurements from the selected JACoP output.
12. Save:
 - the selected colocalization measurements in CSV and Excel format
 - the raw JACoP output
 - the selected fluorogram
 - the variant used and its graphical quality-control parameters.
13. IF results from both VAR0 and VAR1 are available:
 - additionally save separate measurement tables for both variants.

Algorithm 5. Output generation and organization

INPUT:

Analysis outputs generated for each input image

1. Create a dedicated output directory for each analyzed image.
2. Save the preprocessed C1 and C2 images.
3. Save the C1 and C2 fluorescence-intensity histogram data.
4. Export fluorescence-intensity data separately for C1 and C2 in tabular form.
5. Generate a two-panel fluorescence-intensity distribution plot containing the C1 and C2 intensity histograms.

6. For each C2 thresholding method save:
 - boundary-excluded particle-level measurements
 - boundary-excluded segmentation overlay
 - boundary-inclusive particle-level measurements
 - boundary-inclusive segmentation overlay.
7. Generate per-method C2 summary statistics for:
 - particle area
 - perimeter-derived diameter
 - area-equivalent circular diameter
 - particle count.
8. Generate summary tables combining C2 results obtained with the different thresholding methods.
9. Generate a comparison grid containing the preprocessed C2 image and C2 segmentation results obtained using the different automatic thresholding methods.
10. Generate diagnostic plots comparing C2 particle measurements obtained with the different thresholding methods.
11. Export the results of the Shewhart-inspired threshold-method consistency analysis.
12. Save colocalization outputs including:
 - merged two-channel image
 - colocalization measurement tables
 - raw colocalization output
 - selected fluorogram
 - available alternative fluorogram and variant-specific tables.
13. Organize intermediate and output files into dedicated subdirectories for:
 - boundary-inclusive segmentation images
 - boundary-inclusive CSV files
 - thresholding and colocalization CSV files
 - Fiji/ImageJ macro files.
14. Apply the configured filename-shortening rules to the output directory and its files.
15. Complete processing of the current image and proceed to the next designated input image.